Abinav S

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Portfolio **LinkedIn** **GitHub**

PROFESSIONAL SUMMARY

Data Engineer with 3 years of hands-on experience building, optimizing, and maintaining **production data pipelines** using **Python, SQL, and PySpark** for financial data workloads at **Bank of Montreal (BMO) through Virtusa**. Proven track record of improving **pipeline reliability**, optimizing **data processing performance**, reducing **compute costs**, and resolving **production issues**.

WORK EXPERIENCE

Data Engineer

Virtusa – Client: BMO (Bank of Montreal), Chennai, India ***Jan 2024 – Present***

- Built and maintained Python, SQL, and **AWS Glue (PySpark) batch pipelines** for financial datasets with **50M+ records**, using **SQL deduplication, incremental loads, and Redshift COPY** across raw and reporting layers for downstream reporting.
- Orchestrated **Airflow DAGs** across **150+ daily production runs**, implementing business-day logic, conditional branching, and task dependencies.
- Since Jun 2026, have led a **four-person offshore L3 team** owning reliability of BMO's Airflow, Glue and Redshift ETL pipelines; root-caused recurring failures and drove **4+ permanent fixes**: DAG run-concurrency limit, retries with backoff, data type casting, and notification date handling.
- Reduced Redshift query execution time by **~30%** by recommending automated **ANALYZE and VACUUM** maintenance on **~30 high-churn tables**, implemented by the infrastructure team as scheduled **Airflow DAGs**.
- Optimized **Amazon Redshift stored procedures** by replacing **function-based date filters with range predicates**, simplifying deduplication using **QUALIFY**.
- Lowered AWS Glue compute costs by an **estimated $450/month** through shorter job runtimes.
- Added **pre-execution data quality checks to Airflow DAGs**, validating schemas, dates, required columns, and duplicates before starting Glue jobs to catch invalid inputs before processing and downstream delivery.
- Collaborated with **cross-functional development and infrastructure teams** to troubleshoot **ETL bottlenecks**, implement **pipeline improvements**, and ensure **reliable, timely delivery of financial data** to downstream reporting systems.

Software Engineering Intern

Virtusa, Chennai, India ***Jan 2023 – Apr 2023***

- Created a **Python and SQL ETL proof of concept** covering extraction, transformation, validation, and loading.
- Configured **Airflow dependencies and retries**, integrated S3 and Lambda, and containerized the pipeline using **Docker**.

TECHNICAL SKILLS

Languages: Python, SQL, PySpark, Spark SQL
Big Data & Processing: Apache Spark, PySpark, Spark Structured Streaming, Batch Processing, Incremental Loads, Data Deduplication, Schema Validation, Data Quality Checks
Orchestration & ELT: Apache Airflow, AWS Glue, dbt
Cloud Platforms: AWS (S3, Glue, Redshift, Lambda, SNS, IAM, CloudWatch, ECR, DynamoDB); Azure (Data Factory, ADLS Gen2, Synapse Analytics, Databricks)
Data Warehousing & Modeling: Amazon Redshift, Snowflake, PostgreSQL, Dimensional Modeling, Fact and Dimension Tables, Surrogate Keys, Incremental Models, Redshift COPY, Partitioned Parquet
Streaming & Messaging: Apache Kafka, Spark Structured Streaming, Checkpointing, Event Deduplication, Late-Arriving Events
Tools, DevOps & Monitoring: Git, GitHub, Docker, CI/CD, Prometheus, Grafana, Power BI

PROJECTS

Retail Sales Lakehouse

Incremental Batch Data Pipeline

- Built an **Airflow-orchestrated PostgreSQL/API ingestion pipeline** across six entities (**500K+ records**) using watermark-based incremental extraction, input validation, and **six parallel Bronze jobs** feeding Silver processing.
- Engineered **PySpark transformations** across Bronze and Silver with window-based deduplication, schema enforcement, FX enrichment, and row-level reject handling, writing **batch-date partitioned Parquet** across Raw, Bronze, Silver, and Reject zones for **idempotent, rerunnable processing**.
- Loaded **Snowflake** via internal stages and **COPY INTO**, and built **dbt dimensional, fact, and analytical models** with surrogate keys, latest-record handling, and unique, not-null, and relationship tests.

CrickX

Simulated Cricket Streaming Pipeline

- Built and deployed a **Kafka, PySpark Structured Streaming, and PostgreSQL** pipeline, with Streamlit serving ball-by-ball scorecards, player and team statistics, and match analytics across six simulated cricket fixtures.
- Implemented **revision-aware deduplication** and transactional handling of late and corrected events in PostgreSQL; added Spark checkpoints, Prometheus/Grafana monitoring, and Python/PostgreSQL/Spark tests.

CERTIFICATIONS

AWS Certified Data Engineer – Associate 2024
AWS Certified Solutions Architect – Associate 2023
AWS Certified AI Practitioner 2025

EDUCATION

Sri Ramakrishna Engineering College **Coimbatore, India**

Bachelor of Engineering (B.E.), Computer Science – CGPA: 8.0 *May 2023*